Schauder理论（内估计）*

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♣ 回顾Laplace算子的Schauder内估计及给出插值不等式.

对于Poisson方程，我们已经有$C^{2,\alpha}$估计，即Poisson方程的Schauder内估计.即

定理 1 设B_1为$\mathbb{R}^n$中的单位球，$f \in C^\alpha(B_1)$（$\alpha \in (0,1)$），且$|f|_{C^\alpha(B_1)} < \infty$.设$u \in L^\infty(B_1) \cap C^2(B_1)$满足$\Delta u = f$.则$u \in C^{2,\alpha}(B_1)$.进一步，有估计：

$$|u|_{C^{2,\alpha}(B_{1/2})} \leq C(|u|_{L^\infty(B_1)} + |f|_{C^\alpha(B_1)}).$$

其中$C \sim n, \alpha$.

证明 见[1]. □

对单位球进行伸缩平移，得到以x_0为圆心，R为半径的球，我们也有相应的Schauder 内估计.即

定理 2 设$B_R(x_0)$为$\mathbb{R}^n$中半径为R的球，$f \in C^\alpha(B_R(x_0))$（$\alpha \in (0,1)$），且$|f|_{C^\alpha(B_R(x_0))}$.设$u \in L^\infty(B_R(x_0)) \cap C^2(B_R(x_0))$满足$\Delta u = f$.则$u \in C^{2,\alpha}(B_R(x_0))$.进一步，有估计：

$$\begin{aligned}&R|Du|_{L^\infty(B_{R/2}(x_0))} + R^2|D^2u|_{L^\infty(B_{R/2}(x_0))} + R^{2+\alpha}[D^2u]_{C^\alpha(B_{R/2(x_0)})}\\ \leq& C(|u|_{L^\infty(B_R)(x_0)} + R^2|f|_{L^\infty(B_R)(x_0)} + R^{2+\alpha}[f]_{C^\alpha(B_R)(x_0)}).\end{aligned}$$

其中$C \sim n, \alpha$.

给出插值不等式.

引理 1 设$\alpha \in (0,1)$，B_R为$\mathbb{R}^n$中半径为R的球，则对任意的$u \in C^{2,\alpha}(\overline{B}_R)$，有不等式：

$$\sum_{i=1}^{2} R^i|D^iu|_{L^\infty(B_R)} + \sum_{i=0}^{1} R^{i+\alpha}[D^iu]_{C^\alpha(B_R)} \leq C(R^{2+\alpha}[D^2u]_{C^\alpha(B_R)} + |u|_{L^\infty(B_R)}).$$

证明 见[1]. □

注记 1 将不等式写开有

$$R|Du|_{L^\infty(B_R)} + R^2|D^2u|_{L^\infty(B_R)} + R^\alpha[u]_{C^\alpha(B_R)} + R^{1+\alpha}[Du]_{C^\alpha(B_R)} \leq C(R^{2+\alpha}[D^2u]_{C^\alpha(B_R)} + |u|_{L^\infty(B_R)}).$$

可以看出关于R的最低次与最高次可以控制住中间项.

*椭圆方程讨论班

设B_R为$\mathbb{R}^n$中的球，则定义

$$|u|^*_{C^{2,\alpha}(B_R)} = |u|_{L^\infty(B_R)} + R|Du|_{L^\infty(B_R)} + R^2|D^2u|_{L^\infty(B_R)} + R^{2+\alpha}[D^2u]_{C^\alpha(B_R)}.$$

为u的伸缩Hölder模.

♣ 常系数椭圆算子的Schauder内估计.

引理 2 设$f \in C^\alpha(B_R)$（$\alpha \in (0,1)$），(a_{ij})为$n \times n$常对称矩阵，满足：

$$\lambda|\xi|^2 \le a_{ij}\xi_i\xi_j \le \Lambda|\xi|^2 \quad \forall \xi \in \mathbb{R}^n.$$

设$u \in C^2(B_R)$满足$a_{ij}D_{ij}u = f$，则$u \in C^{2,\alpha}(B_R)$.进一步，有估计：

$$|u|^*_{C^{2,\alpha}(B_{R/2})} \le C(|u|_{L^\infty(B_R)} + R^2|f|^*_{C^\alpha(B_R)}).$$

证明

- $a_{ij} = \delta_{ij}$，即$a_{ij}D_{ij}u = f$为$\Delta u = f$，从而已证，即**定理1**.
- 一般情形.

<u>基本思路：找一个线性变换将一般常系数椭圆方程变为Poisson方程，然后分析线性变换后的区域问题.</u>

设$A = (a_{ij})$，由于A为对称矩阵，则存在正交矩阵P，使得

$$P'AP = \mathrm{diag}\{\lambda_1, \lambda_2, \cdots, \lambda_n\}.$$

令$D = \mathrm{diag}\{\lambda_1^{-1/2}, \lambda_2^{-1/2}, \cdots, \lambda_n^{-1/2}\}$，则

$$D'P'APD = I_n \text{（单位矩阵）}.$$

令$Q = PD$，即$Q'AQ = I_n$.令$y = Qx$，则$x = Q^{-1}y$.令

$$\tilde{u}(y) = u(x) = u(Q^{-1}y), \quad \tilde{f}(y) = f(x) = f(Q^{-1}y).$$

计算$\Delta\tilde{u}(y)$，即

$$\begin{aligned}
\Delta\tilde{u}(y) &= \sum_{k=1}^n \frac{\partial^2\tilde{u}(y)}{\partial y_k^2} = \sum_{k=1}^n \left(\sum_{i,j=1}^n \frac{\partial^2 u(x)}{\partial x_i \partial x_j} \cdot \frac{\partial x_i}{\partial y_k} \cdot \frac{\partial x_j}{\partial y_k} \right) \\
&= \sum_{i,j=1}^n \left(\frac{\partial^2 u}{\partial x_i \partial x_j} \cdot \sum_{k=1}^n \frac{\partial x_i}{\partial y_k} \cdot \frac{\partial x_j}{\partial y_k} \right) \\
&= \mathrm{tr}(D^2u \cdot (Q^{-1})' \cdot (Q^{-1})) = \mathrm{tr}(A \cdot D^2u) \\
&= \sum_{i,j=1}^n a_{ij}D_{ij}u = f(x) = \tilde{f}(y).
\end{aligned}$$

即$\tilde{u}(y)$满足Poisson方程.

现在分析区域问题.

由于P为正交矩阵，则P作用在向量上，保持向量的长度不变，从而

$$|Qx| = |PDx| = |Dx| = \left(\sum_{i=1}^n \lambda_i^{-1}x_i^2 \right)^{1/2}.$$

由定理的条件，知

$$\Lambda^{-1/2}|x| \le |Qx| \le \lambda^{-1/2}|x|.$$

任取$x \in B_{R/2}$，$x' \in \partial B_R$，则

$$|Q(x-x')| \ge \Lambda^{-1/2}|x-x'| \ge \frac{R}{2}\Lambda^{-1/2}.$$

由于线性变换将凸区域的边界映为边界，则

$$\mathrm{dist}(Q(B_{R/2}), Q(\partial B_R)) \ge \frac{R}{2}\Lambda^{-1/2}.$$

取$\tau = \min\{\frac{1}{2}\Lambda^{-1/2}, \frac{1}{2}\} \in (0,1)$.显然

$$Q(\overline{B}_{R/2}) \subset \bigcup_{y_i \in Q(B_{R/2})} B_{\tau R/2}(y_i),$$

从而存在有限个子覆盖使得

$$Q(B_{R/2}) \subset \bigcup_{i=1}^{N} B_{\tau R/2}(y_i) \subset \bigcup_{i=1}^{N} B_{\tau R}(y_i).$$

由于$\mathrm{dist}(y_i, Q(\partial B_R)) \ge \frac{R}{2}\Lambda^{-1/2}$（$\forall i = 1, 2, \cdots, N$），则

$$B_{\tau R}(y_i) \subset Q(B_R),$$

从而

$$Q(B_{R/2}) \subset \bigcup_{i=1}^{N} B_{\tau R/2}(y_i) \subset \bigcup_{i=1}^{N} B_{\tau R}(y_i) \subset Q(B_R).$$

现在在区域$B_{\tau R}(y_i)$上应用Laplace算子的Schauder估计，即

$$|\tilde{u}|^*_{C^{2,\alpha}(B_{\tau R/2})} \le C(|\tilde{u}|_{L^\infty(B_{\tau R})} + R^2|\tilde{f}|_{C^\alpha(B_{\tau R})}).$$

从而

$$\begin{aligned}
|\tilde{u}|^*_{C^{2,\alpha}(Q(B_{R/2}))} &\le \sum_{i=1}^{N} |\tilde{u}|^*_{C^{2,\alpha}(B_{\tau R/2}(y_i))} \\
&\le \sum_{i=1}^{N} \Big(C(|\tilde{u}|_{L^\infty(B_{\tau R}(y_i))} + R^2|\tilde{f}|_{C^\alpha(B_{\tau R}(y_i))}) \Big) \\
&\le C(|\tilde{u}|_{L^\infty(Q(B_R))} + R^2|\tilde{f}|_{C^\alpha(Q(B_R))}).
\end{aligned}$$

故

$$|u|^*_{C^{2,\alpha}(B_{R/2})} \le C(|u|_{L^\infty(B_R)} + R^2|f|^*_{C^\alpha(B_R)}).$$

□

注记 2 这里的后半部分证明应该能够简化.

对于*Poisson*方程的*Schauder*内估计，我们有如下事实，即一般区域上的*Schauder*内估计.

事实 *1* 设$\Omega \subset \mathbb{R}^n$为有界区域.$f \in C^\alpha(\Omega)$（$\alpha \in (0,1)$），且$\| f \|_{C^\alpha(\Omega)} < \infty$.若$u \in L^\infty(\Omega) \cap C^2(\Omega)$满足方程

$$\Delta u = f \quad in\ \Omega.$$

则$u \in C^{2,\alpha}(\Omega)$，且

$$\| u \|_{C^{2,\alpha}(\Omega')} \leq C(\| u \|_{L^\infty(\Omega)} + \| f \|_{C^\alpha(\Omega)}).$$

其中$c \sim n, \alpha, \mathrm{dist}(\Omega', \partial\Omega)$.

♣ **主系数振幅足够小的Schauder内估计.**

引理 3 $B_R \subset \mathbb{R}^n$，L为B_R上的椭圆算子.设a_{ij}, b_i, c为B_R上的C^α-函数（$\alpha \in (0,1)$）.满足$a_{ij}\xi_i\xi_j \geq \lambda|\xi|^2$（$\forall x \in B_R, \xi \in \mathbb{R}^n$），且

$$|a_{ij}|^*_{C^\alpha(B_R)} + R|b_i|^*_{C^\alpha(B_R)} + R^2|c|^*_{C^\alpha(B_R)} \leq \Lambda.$$

设$u \in C^{2,\alpha}(B_R)$满足

$$Lu = f \quad in\ B_R.$$

存在$\varepsilon \sim n, \alpha, \lambda, \Lambda$，使得

$$\sup_{x,y \in B_R} |a_{ij}(x) - a_{ij}(y)| \leq \varepsilon.$$

则有估计

$$|u|^*_{C^{2,\alpha}(B_{R/2})} \leq C(|u|_{L^\infty(B_R)} + R^2|f|^*_{C^\alpha(B_R)}).$$

证明 根据插值不等式及要证明的内容知道，我们只需证明

$$R^{2+\alpha}[D^2u]_{C^\alpha(B_R)} \leq C(|u|_{L^\infty(B_R)} + R^2|f|_{L^\infty(B_R)} + R^{2+\alpha}[f]_{C^\alpha(B_R)}).$$

任取$B_r \subset B_R$（$r \in (0, R]$）.令

$$a_{ij}(0)D_{ij}u = \tilde{f} = f - cu - b_iD_iu - (a_{ij} - a_{ij}(0))D_{ij}u.$$

在B_r中，根据**引理2**有估计：

$$r^{2+\alpha}[D^2u]_{C^\alpha(B_{r/2})} \leq C(|u|_{L^\infty(B_r)} + r^2|f|_{L^\infty(B_r)} + r^{2+\alpha}[f]_{C^\alpha(B_r)}).$$

分别进行估计：

- $r^2|f|_{L^\infty(B_r)}$.

$$\begin{aligned}
&r^2|\tilde{f}|_{L^\infty(B_r)} \\
\leq &r^2|f|_{L^\infty(B_r)} + r^2|c|_{L^\infty(B_r)}|u|_{L^\infty(B_r)} + r^2|b_i|_{L^\infty(B_r)}|Du|_{L^\infty(B_r)} + r^2|a_{ij} - a_{ij}(0)|_{L^\infty(B_r)}|D^2u|_{L^\infty(B_r)},
\end{aligned}$$

- $r^{2+\alpha}[\tilde{f}]_{C^\alpha(B_r)}$.

$$\begin{aligned}
r^{2+\alpha}[\tilde{f}]_{C^\alpha(B_r)} \leq &r^{2+\alpha}[f]_{C^\alpha(B_r)} + r^{2+\alpha}[c]_{C^\alpha(B_r)}|u|_{L^\infty(B_r)} + r^{2+\alpha}|c|_{L^\infty(B_r)}[u]_{C^\alpha(B_r)} \\
&+ r^{2+\alpha}[b_i]_{C^\alpha(B_r)}|Du|_{L^\infty(B_r)} + r^{2+\alpha}|b_i|_{L^\infty(B_r)}[Du]_{C^\alpha(B_r)} \\
&+ r^{2+\alpha}|a_{ij} - a_{ij}(0)|_{L^\infty(B_r)}[D^2u]_{C^\alpha(B_r)} + r^{2+\alpha}[a_{ij} - a_{ij}(0)]_{C^\alpha(B_r)}|D^2u|_{L^\infty(B_r)}.
\end{aligned}$$

其中我们应用了如下事实，

$$\begin{aligned}
[cu]_{C^\alpha(B_r)} &= \sup_{x,y \in B_r} \frac{|c(x)u(x) - c(y)u(y)|}{|x-y|^\alpha} \\
&= \sup_{x,y \in B_r} \frac{c(x)u(x) - c(x)u(y) + c(x)u(y) - c(y)u(y)}{|x-y|^\alpha} \\
&\leq \sup_{x,y \in B_r} \frac{|c(x)(u(x) - u(y))| + |u(y)(c(x) - c(y))|}{|x-y|^\alpha}
\end{aligned}$$

$$
\begin{aligned}
&\leq \sup_{x,y\in B_r}|c(x)|\frac{|u(x)-u(y)|}{|x-y|^\alpha}+\sup_{x,y\in B_r}|u(y)|\frac{|c(x)-c(y)|}{|x-y|^\alpha}\\
&\leq |c|_{L^\infty(B_r)}[u]_{C^\alpha(B_r)}+|u|_{L^\infty(B_r)}[c]_{C^\alpha(B_r)}.
\end{aligned}
$$

由定理中的条件：

$$
|a_{ij}|_{L^\infty(B_r)}+r^\alpha[a_{ij}]_{C^\alpha(B_r)}+r|b_i|_{L^\infty(B_r)}+r^{1+\alpha}[b_i]_{C^\alpha(B_r)}+r^2|c|_{L^\infty(B_r)}+r^{2+\alpha}[c]_{C^\alpha(B_r)}\leq\Lambda,
$$

则

$$
\begin{aligned}
r^{2+\alpha}[D^2u]_{C^\alpha(B_{r/2})}\leq&C(|u|_{L^\infty(B_r)}+r^2|f|_{L^\infty(B_r)}+r^{2+\alpha}[f]_{C^\alpha(B_r)}\\
&+r^2(|a_{ij}|_{L^\infty(B_r)}+r^\alpha[a_{ij}]_{C^\alpha(B_r)})|D^2u|_{L^\infty(B_r)}\\
&+r(r|b_i|_{L^\infty(B_r)}+r^{1+\alpha}[b_i]_{C^\alpha(B_r)})|Du|_{L^\infty(B_r)}\\
&+(r^2|c|_{L^\infty(B_r)}+r^{2+\alpha}[c]_{C^\alpha(B_r)})|u|_{L^\infty(B_r)}\\
&+r^\alpha[u]_{C^\alpha(B_r)}+r^{1+\alpha}[Du]_{C^\alpha(B_r)}+r^{2+\alpha}|a_{ij}-a_{ij}(0)|_{L^\infty(B_r)}[D^2u]_{C^\alpha(B_r)})\\
\leq&C(r^2|f|_{L^\infty(B_r)}+r^{2+\alpha}[f]_{C^\alpha(B_r)}+|u|_{L^\infty(B_r)}+r^\alpha[u]_{C^\alpha(B_r)}\\
&+r|Du|_{L^\infty(B_r)}+r^{1+\alpha}[Du]_{C^\alpha(B_r)}+r^2|D^2u|_{L^\infty(B_r)}\\
&+|a_{ij}-a_{ij}(0)|_{L^\infty(B_r)}r^{2+\alpha}[D^2u]_{C^\alpha(B_r)}).
\end{aligned}
$$

由插值不等式

$$
\begin{aligned}
&|u|_{L^\infty(B_r)}+r|Du|_{L^\infty(B_r)}+r^2|D^2u|_{L^\infty(B_r)}+r^\alpha[u]_{C^\alpha(B_r)}+r^{1+\alpha}[Du]_{C^\alpha(B_r)}\\
\leq&C(r^{2+\alpha}[D^2u]_{C^\alpha(B_r)}+|u|_{L^\infty(B_r)})\leq\varepsilon r^{2+\alpha}[D^2u]_{C^\alpha(B_r)}+C(\varepsilon)|u|_{L^\infty(B_r)}.
\end{aligned}
$$

从而我们得到估计：

$$
r^{2+\alpha}[D^2u]_{C^\alpha(B_{r/2})}\leq C(r^2|f|_{L^\infty(B_r)}+r^{2+\alpha}[f]_{C^\alpha(B_r)}+\varepsilon r^{2+\alpha}[D^2u]_{C^\alpha(B_r)}+C(\varepsilon)|u|_{L^\infty(B_r)}).
$$

$\Longrightarrow$

$$
r^{2+\alpha}[D^2u]_{C^\alpha(B_{r/2}(x_0))}\leq C(R^2|f|_{L^\infty(B_R)}+R^{2+\alpha}[f]_{C^\alpha(B_R)})+C\varepsilon r^{2+\alpha}[D^2u]_{C^\alpha(B_r(x_0))}+C(\varepsilon)|u|_{L^\infty(B_R)}.(*)
$$

由于上式左右端区域不同，则不等号右边第三项不管ε取多小，都不能被左端吸收，下面将对区域进行分析.令

$$
Q=\sup\{r^{2+\alpha}[D^2u]_{C^\alpha(B_{r/2}(x_0))}\mid B_r(x_0)\subset B_R\}.
$$

显然有

$$
r^{2+\alpha}[D^2u]_{C^\alpha(B_{r/4}(x_0))}\leq C(R^2|f|_{L^\infty(B_R)}+R^{2+\alpha}[f]_{C^\alpha(B_R)})+C\varepsilon Q+C_\varepsilon|u|_{L^\infty(B_R)}.
$$

任取$B_r(x_0)\subset B_R$，则存在$z_1,z_2,\cdots,z_N$，使得

$$
B_{r/2}(x_0)\subset\bigcup_{i=1}^N B_{r/16}(z_i),\quad \text{且 } B_{r/2}(z_i)\subset B_R, i=1,2,\cdots,N.
$$

对任意的$B_{r/2}(z_i)\subset B_R$，则

$$
r^{2+\alpha}[D^2u]_{C^\alpha(B_{r/4}(z_i))}\leq Q.
$$

对$B_{r/4}(z_i)$应用（$*$），有

$$r^{2+\alpha}[D^2u]_{C^\alpha(B_{r/8}(z_i))} \le C(R^2|f|_{L^\infty(B_R)} + R^{2+\alpha}[f]_{C^\alpha(B_R)}) + C\varepsilon Q + C(\varepsilon)|u|_{L^\infty(B_R)}.$$

下面说明

$$[D^2u]_{C^\alpha(B_{r/2}(x_0))} \le C \cdot \sum_{i=1}^{N}[D^2u]_{C^\alpha(B_{r/8}(z_i))}.$$

取$x_1, x_2 \in B_{r/2}(x_0)$.
设$x_1 \in B_{r/16}(z_1)$.
• $x_2 \in B_{r/8}(z_1)$.则

$$\frac{|D^2u(x_1) - D^2u(x_2)|}{|x_1 - x_2|^\alpha} \le [D^2u]_{C^\alpha(B_{r/8}(z_1))}.$$

• $x_2 \notin B_{r/8}(z_1)$.则$|x_1 - x_2| > \dfrac{r}{16}$.设$x_2 \in B_{r/16}(z_2)$，则

$$|D^2u(x_1) - D^2u(x_2)| \le |D^2u(x_1) - D^2u(z_1)| + |D^2u(z_1) - D^2u(z_2)| + |D^2u(z_2) - D^2u(x_2)|.$$

对于z_1, z_2，我们可以找到一列z_{k_i}，即$z_{k_0} = z_1, z_{k_1}, \cdots, z_{k_l} = z_2$，其中$|z_{k_{i+1}} - z_{k_i}| < \dfrac{r}{8}$，则

$$|D^2u(x_1) - D^2u(x_2)| \le \left(\frac{r}{8}\right)^\alpha \cdot \sum_i [D^2u]_{C^\alpha(B_{r/8}(z_i))}.$$

故

$$[D^2u]_{C^\alpha(B_{r/2}(x_0))} \le C \cdot \sum_{i=1}^{N}[D^2u]_{C^\alpha(B_{r/8}(z_i))}.$$

从而就有估计：

$$r^{2+\alpha}[D^2u]_{C^\alpha(B_{r/2}(x_0))} \le CN(R^2|f|_{L^\infty(B_R)} + R^{2+\alpha}[f]_{C^\alpha(B_R)}) + CN\varepsilon + C(\varepsilon)N|u|_{L^\infty(B_R)}.$$

同时取上确界，且取$CN\varepsilon < \dfrac{1}{2}$，则有

$$Q \le C(R^2|f|_{L^\infty(B_R)} + R^{2+\alpha}[f]_{C^\alpha(B_R)} + |u|_{L^\infty(B_R)}).$$

根据Q的定义，令$r \longrightarrow R$，则有

$$R^{2+\alpha}[D^2u]_{C^\alpha(B_{R/2})}C(R^2|f|_{L^\infty(B_R)} + R^{2+\alpha}[f]_{C^\alpha(B_R)} + |u|_{L^\infty(B_R)}).$$

□

♣ 一致椭圆算子的Schauder内估计.

定理 3 $B_R \subset \mathbb{R}^n$，L为B_R上的椭圆算子.设a_{ij}, b_i, c为B_R上的C^α-函数（$\alpha \in (0,1)$）.满足$a_{ij}\xi_i\xi_j \ge \lambda|\xi|^2$（$\forall x \in B_R, \xi \in \mathbb{R}^n$），且

$$|a_{ij}|^*_{C^\alpha(B_R)} + R|b_i|^*_{C^\alpha(B_R)} + R^2|c|^*_{C^\alpha(B_R)} \le \Lambda.$$

设$u \in C^{2,\alpha}(B_R)$满足

$$Lu = f \quad in\ B_R.$$

则有估计

$$|u|^*_{C^{2,\alpha}(B_{R/2})} \le C(|u|_{L^\infty(B_R)} + R^2|f|^*_{C^\alpha(B_R)}).$$

证明 对任意的$B_{R_0}(x_0)\subset B_R$，有

$$\sup_{x,y\in B_{R_0}(x_0)}|a_{ij}(x)-a_{ij}(y)|\le R_0^{\alpha}[a_{ij}]_{C^{\alpha}(B_{R_0}(x_0))}\le\left(\frac{R_0}{R}\right)^{\alpha}R^{\alpha}[a_{ij}]_{C^{\alpha}(B_R)}\le\left(\frac{R_0}{R}\right)^{\alpha}\Lambda.$$

取$R_0<R$，使得

$$\left(\frac{R_0}{R}\right)^{\alpha}\Lambda=\varepsilon.$$

由于此时主系数的振幅足够小，则应用**引理3**，有

$$\begin{aligned}&R_0^{2+\alpha}[D^2u]_{C^{\alpha}(B_{R_0/2}(x_0))}\\ \le&C\left(R_0^2|f|_{L^{\infty}(B_R(x_0))}+R_0^{2+\alpha}[f]_{C^{\alpha}(B_R(x_0))}+|u|_{L^{\infty}(B_{R_0}(x_0))}\right)\\ \le&C\left(R^2|f|_{L^{\infty}(B_R)}+R^{2+\alpha}[f]_{C^{\alpha}(B_R)}+|u|_{L^{\infty}(B_R)}\right).\end{aligned}$$

现在需要将不等号左端的项进行调整.
任取$x,y\in B_{R/2}$.

- $|x-y|\le R_0$.取$x_0=\dfrac{x+y}{2}$，显然

$$R_0^{2+\alpha}\frac{|D^2u(x)-D^2u(y)|}{|x-y|^{\alpha}}\le C\left(R^2|f|_{L^{\infty}(B_R)}+R^{2+\alpha}[f]_{C^{\alpha}(B_R)}+|u|_{L^{\infty}(B_R)}\right).$$

- $|x-y|\ge R_0$.则

$$R_0^{2+\alpha}\frac{|D^2u(x)-D^2u(y)|}{|x-y|^{\alpha}}\le 2R_0^2|D^2u|_{L^{\infty}(B_{R/2})}.$$

从而

$$R_0^{2+\alpha}[D^2u]_{C^{\alpha}(B_{R/2})}\le C\left(R^2|f|_{L^{\infty}(B_R)}+R^{2+\alpha}[f]_{C^{\alpha}(B_R)}+|u|_{L^{\infty}(B_R)}\right)+2R_0^2|D^2u|_{L^{\infty}(B_{R/2})}.$$

由于$\dfrac{R}{R_0}$为固定的值，则

$$R^{2+\alpha}[D^2u]_{C^{\alpha}(B_{R/2})}\le C\left(R^2|f|_{L^{\infty}(B_R)}+R^{2+\alpha}[f]_{C^{\alpha}(B_R)}+|u|_{L^{\infty}(B_R)}+R^2|D^2u|_{L^{\infty}(B_{R/2})}\right).$$

对上式最后一项使用插值不等式，即

$$R^2|D^2u|_{L^{\infty}(B_{R/2})}\le\varepsilon R^{2+\alpha}[D^2u]_{C^{\alpha}(B_{R/2})}+C_{\varepsilon}|u|_{L^{\infty}(B_R)}.$$

取ε足够小，从而

$$R^{2+\alpha}[D^2u]_{C^{\alpha}(B_{R/2})}\le C\left(R^2|f|_{L^{\infty}(B_R)}+R^{2+\alpha}[f]_{C^{\alpha}(B_R)}+|u|_{L^{\infty}(B_R)}\right).$$

□

♣ **一般区域上的Schauder内估计.**

定理 4 设Ω为$\mathbb{R}^n$上的有界区域，L为Ω上的椭圆算子.设a_{ij},b_i,c为Ω上的C^{α}-函数（$\alpha\in(0,1)$）.满足

$$a_{ij}\xi_i\xi_j\ge\lambda|\xi|^2,\quad\forall x\in\Omega,\xi\in\mathbb{R}^n.$$

且有

$$|a_{ij}|_{C^{\alpha}(\Omega)}+|b_i|_{C^{\alpha}(\Omega)}+|c|_{C^{\alpha}(\Omega)}\le\Lambda.$$

设$u \in L^\infty(\Omega) \cap C^{2,\alpha}(\Omega)$为

$$Lu = f \quad in\ \Omega$$

的解，其中$f \in C^\alpha(\Omega)$，$|f|_{C^\alpha(\Omega)} < \infty$.则对任意的$\Omega' \subset\subset \Omega$，

$$|u|_{C^{2,\alpha}(\Omega')} \le C(|u|_{L^\infty(\Omega)} + |f|_{C^\alpha(\Omega)}),$$

其中$C \sim n, \alpha, \lambda, \Lambda, \Omega', \mathrm{dist}(\Omega', \partial\Omega)$.

证明 令$d = \mathrm{dist}(\Omega', \partial\Omega)$，由于$\Omega'$为$\mathbb{R}^n$上的有界闭集，则存在$x_1, \cdots, x_N \in \Omega'$，使得

$$\Omega' \subset \bigcup_{i=1}^N B_{d/2}(x_i), \quad B_d(x_i) \subset \Omega.$$

在$B_d(x_i)$上应用圆上的Schauder内估计，有

$$\begin{aligned}|u|^*_{C^{2,\alpha}(B_{d/2(x_i)})} &\le C(|u|_{L^\infty(B_d(x_i))} + d^2|f|_{L^\infty(B_d(x_i))} + d^{2+\alpha}[f]_{C^\alpha(B_d(x_i))}) \\ &\le C(|u|_{L^\infty(\Omega)} + |f|_{L^\infty(\Omega)} + [f]_{C^\alpha(\Omega)}).\end{aligned}$$

从而有估计：

$$\begin{aligned}|u|_{C^{2,\alpha}(\Omega')} &\le \sum_{i=1}^N |u|^*_{C^{2,\alpha}(B_{d/2}(x_i))} \\ &\le C(|u|_{L^\infty(\Omega)} + |f|_{L^\infty(\Omega)} + [f]_{C^\alpha(\Omega)}).\end{aligned}$$

■ **另证.**
我们只需证明

$$d\sup_{\Omega'}|Du| + d^2\sup_{\Omega'}|D^2u| + d^{2+\alpha}[D^2u]_{C^\alpha(\Omega')} \le C\left(\sup_\Omega|u| + d^2\sup_\Omega|f| + d^{2+\alpha}[f]_{C^\alpha(\Omega)}\right).$$

任取$x \in \Omega'$，则$B_d(x) \subset \Omega$.
• $|x-y| < \dfrac{d}{2}$.
在$B_d(x)$上应用球上的Schauder内估计，有

$$\begin{aligned}&d|Du(x)| + d^2|D^2u(x)| + d^{2+\alpha}\sup_{y\in B_{d/2}(x)}\frac{|D^2u(x)-D^2u(y)|}{|x-y|^\alpha} \\ \le& C\left(\sup_{B_d(x)}|u| + d^2\sup_{B_d(x)}|f| + d^{2+\alpha}[f]_{C^\alpha(B_d(x))}\right) \\ \le& C\left(\sup_\Omega|u| + d^2\sup_\Omega|f| + d^{2+\alpha}[f]_{C^\alpha(\Omega)}\right).\end{aligned}$$

• $|x-y| \ge \dfrac{d}{2}$.则有

$$d^{2+\alpha}\frac{|D^2u(x)-D^2u(y)|}{|x-y|^\alpha} \le 2d^2(|D^2u(x)| + |D^2u(y)|) \le 4d^2\sup_{\Omega'}|D^2u|.$$

□

♣ **一致收敛的函数列的Schauder内估计.**

定理 5 设$\Omega \subset \mathbb{R}^n$为有界区域，$L$为$\Omega$上的椭圆算子.设$a_{ij}, b_i, c$为$\Omega$上的$C^\alpha$-函数（$\alpha \in (0,1)$）.满足

$$a_{ij}\xi_i\xi_j \geq \lambda|\xi|^2, \quad \forall x \in \Omega, \xi \in \mathbb{R}^n.$$

且有

$$|a_{ij}|_{C^\alpha(\Omega)} + |b_i|_{C^\alpha(\Omega)} + |c|_{C^\alpha(\Omega)} \leq \Lambda.$$

设$u_k \in L^\infty(\Omega) \cap C^{2,\alpha}(\Omega)$满足

$$Lu_k = f \quad in\ \Omega.$$

其中$f \in C^\alpha(\Omega)$，$|f|_{C^\alpha(\Omega)} < \infty$.则存在子列$\{u_{k_i}\}_{i=1}^\infty$和$u \in C^{2,\alpha}(\Omega)$，使得$u_{k_i} \longrightarrow u$在$\Omega$的紧子集上一致收敛，且

$$Lu = f \quad in\ \Omega.$$

证明 见[2].知$u_{k_i} \longrightarrow u$在$\Omega$的紧子集上一致收敛.
由于$Lu_{k_i} = f$，则$Lu = f$.取$x, y \in \Omega$，$x \neq y$，有

$$\frac{|D^2u(x) - D^2u(y)|}{|x-y|^\alpha} = \lim_{i\to\infty} \frac{|D^2u_{k_i}(x) - D^2u_{k_i}(y)|}{|x-y|^\alpha} \leq C < \infty,$$

从而

$$u \in C^{2,\alpha}(\Omega).$$

□

参考文献

[1] Qing Han,An Introduction to Elliptic Differential Equations,Preliminary Version.

[2] David Gilbarg,Neil S.Trudinger,Elliptic Partial Differential Equations of Second Order,Springer,1998.